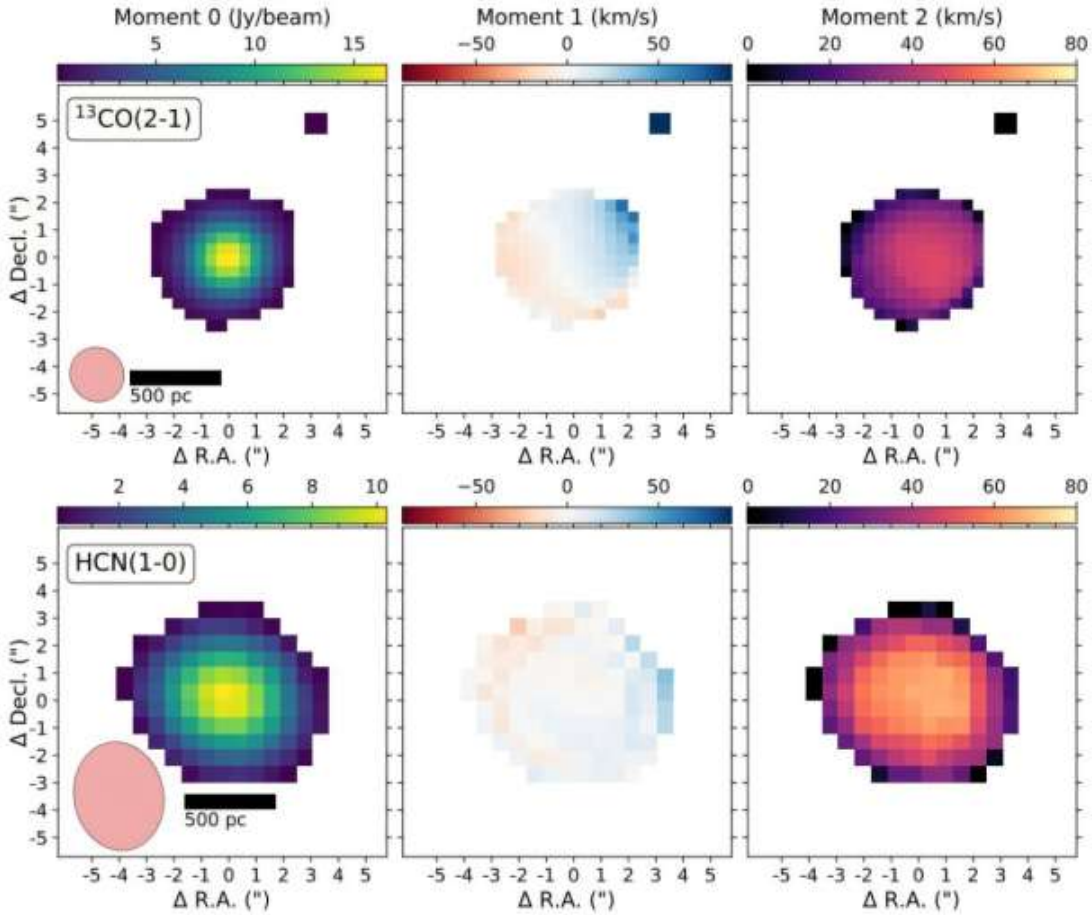
ALMA datasets elucidate nearby galaxy NGC 1266's massive molecular outflow

Tomasz Nowakowski



Moment 0, 1, and 2 maps of the ALMA Cycle 0 observations of NGC 1266. Credit: arXiv (2025). DOI: 10.48550/arxiv.2512.09889

By analyzing the archival data from the Atacama Large Millimeter/submillimeter Array (ALMA), an international team of astronomers has inspected the outflow of a nearby galaxy known as NGC 1266. Results of the new study, presented Dec. 11 on the *arXiv* pre-print server, could help us better understand the nature of this galaxy.

Early-type galaxy with massive molecular outflow

Located some 97.5 million light years away, NGC 1266 (also known as PGC 12131) is a nearby lenticular galaxy in the constellation Eridanus. It is a quenching, early-type galaxy with a half-light radius of 9,650 light years and an estimated mass of about 25 billion solar masses.

Previous observations of NGC 1266 have found that it hosts an obscured active galactic nucleus (AGN) and that it experiences a massive molecular outflow likely driven by this AGN. A group of astronomers led by Justin Atsushi Otter of the Johns Hopkins University in Baltimore, Maryland, decided to take a closer look at this outflow, hoping to unveil its impact on global star formation regulation across the galaxy.

"NGC 1266 is an excellent case study for investigating the outflows and star formation quenching because it is a nearby (D~30 Mpc) AGN host galaxy with an outflow driving shocks through the interstellar medium (ISM) and has recently quenched its star formation outside the nucleus," the scientists explain.

Exploring gas emission

As part of the study, Otter's team combed through the datasets containing ALMA Band 3, 6, and 7 observations of NGC 1266 obtained with the 12-m array in Cycle 0 in the extended configuration. All of these observations were centered on the coordinates of the galaxy's optical center.

The researchers focused on the hydrogen cyanide (HCN) and carbon monoxide (CO) emission in the outflow of NGC 1266. They detected substantial HCN emission in the outflow, with HCN/CO flux ratio consistent with early type galaxies. Moreover, they identified a highly optically thin CO-emitting gas in the outflow with low ^{13}CO abundance.

The astronomers assume that the HCN-emitting gas likely traces a denser phase of the outflow than the diffuse gas traced by optically thin CO emission.

"This denser phase may be concentrated in clumps entrained in the outflow, while the surrounding molecular gas traced by CO is warm, diffuse, and optically thin," the authors of the paper conclude.

Losing all its gas reservoirs

Based on the collected data, the scientists were able to calculate the upper limit of the molecular outflow rate, which was estimated to be 85 solar masses. This value suggests that NGC 1266 will deplete its gas reservoirs in 450 million years or longer.

The scientists conclude that the estimated gas depletion time for NGC 1266 indicates that low-level AGN activity is likely capable of slowly expelling the molecular gas through the post-starburst phase, potentially resulting in long-term quiescence.

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